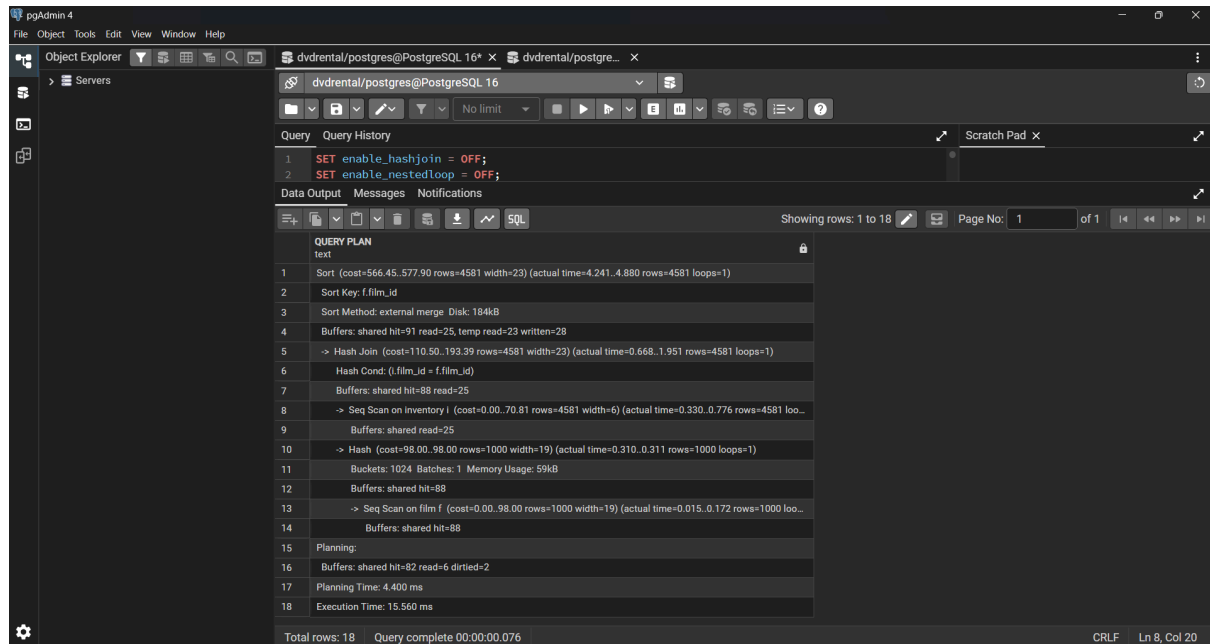


Lab Exercise 3: Triggering Merge Join



The screenshot shows the pgAdmin 4 interface with a query window open. The query is:

```
1 SET enable_hashjoin = OFF;  
2 SET enable_nestedloop = OFF;
```

The query plan shows the following steps:

- Sort (cost=566.45..577.90 rows=4581 width=23) (actual time=4.241..4.880 rows=4581 loops=1)
- Sort Key: f.film_id
- Sort Method: external merge Disk: 184kB
- Buffers: shared hit=91 read=25, temp read=23 written=28
- > Hash Join (cost=110.50..193.39 rows=4581 width=23) (actual time=0.668..1.951 rows=4581 loops=1)
- Hash Cond: (i.film_id = f.film_id)
- Buffers: shared hit=88 read=25
- > Seq Scan on inventory i (cost=0.00..70.81 rows=4581 width=6) (actual time=0.330..0.776 rows=4581 loops=1)
- Buffers: shared read=25
- > Hash (cost=98.00..98.00 rows=1000 width=19) (actual time=0.310..0.311 rows=1000 loops=1)
- Buckets: 1024 Batches: 1 Memory Usage: 59kB
- Buffers: shared hit=88
- > Seq Scan on film f (cost=0.00..98.00 rows=1000 width=19) (actual time=0.015..0.172 rows=1000 loops=1)
- Buffers: shared hit=88
- Planning:
- Buffers: shared hit=82 read=6 dirtied=2
- Planning Time: 4.400 ms
- Execution Time: 15.560 ms

Total rows: 18 Query complete 00:00:00.076

- Structure: Do you see a Sort node? Or did it use Index Scan to get sorted data for free?
→answer: Yes, there is a Sort node at the very top (Line 1 in your query plan). It did not use an Index Scan to get sorted data for free; instead, it used Sequential Scans (Seq Scan), performed a Hash Join (Line 5), and then explicitly sorted the final results.
- Compare: Reset the settings (RESET ALL;) and remove the ORDER BY. Does it switch back to Hash Join? Why?
→answer: Yes, it will default to a Hash Join. When you join two entire tables without explicitly requesting the output to be sorted (ORDER BY), building a hash table in RAM is generally much faster and more efficient for PostgreSQL than forcing a sorting process for a Merge Join.